

ECON 6020 TUTORIAL 1

- By: Teresa Ng
- Lesson One
Date: 24/04/26

Today's Focus:

"Lagrangian"

λ

Example:

$$-\max xy \text{ s.t. } 4x + 3y = 21$$

Steps:

1. Form the lagrangian

2. FOC

3. Find x, y

Solution:

$$\text{Step 1: } xy - \lambda(4x + 3y - 21)$$

Step 2: FOC

$$L'_x: y - 4\lambda = 0$$

$$L'_y: x - 3\lambda = 0$$

$$L'_\lambda: 4x + 3y = 21$$

Step: Solve for x, y

$$y = 4\lambda$$

$$x = 3\lambda \quad 12\lambda + 12\lambda = 21$$

$$24\lambda = 21$$

$$\lambda = \frac{21}{24} = \frac{7}{8}$$

$$y = \frac{7}{2} \quad x = \frac{21}{8}$$

Try a Simple Qns:

$$(a) \min -18Q_1 + Q_1^2 - 2Q_1Q_2 - 4Q_2 + Q_2^2$$

$$\text{s.t } Q_1 + Q_2 = 9$$

$$(b) \max x^{\frac{1}{2}} y^{\frac{1}{2}}$$

$$\text{s.t } 40000x + y = 4000000$$

Solⁿ :

$$-18Q_1 + Q_1^2 - 2Q_1Q_2 - 4Q_2 + Q_2^2 - \lambda(Q_1 + Q_2 - 9)$$

$$L'_1 = -18 + 2Q_1 - 2Q_2 - \lambda = 0$$

$$L'_2 = -2Q_1 - 4 + 2Q_2 - \lambda = 0$$

$$-18 + 2Q_1 - 2Q_2 = -2Q_1 - 4 + 2Q_2$$

$$-14 = -4Q_1 + 4Q_2$$

$$Q_1 + Q_2 = 9$$

$$Q_1 = 9 - Q_2$$

$$-14 = -4(9 - Q_2) + 4Q_2$$

$$-14 = -36 + 4Q_2 + 4Q_2$$

$$18 = 8Q_2$$

$$Q_2 = \frac{18}{8} = \frac{9}{4}$$

$$Q_1 = 9 - \frac{9}{4} = \frac{36}{4} - \frac{9}{4} = \frac{27}{4}$$

$$x^{\frac{1}{2}} y^{\frac{1}{2}} - \lambda (40000x + y - 40000000)$$

$$L'_x = x^{-\frac{1}{2}} y^{\frac{1}{2}} - 40000\lambda = 0$$

$$L'_y = x^{\frac{1}{2}} y^{-\frac{1}{2}} - \lambda = 0$$

$$\frac{y}{x} = 40000$$

$$y = 40000x$$

$$~~80000x = 40000000~~$$

$$8x = 40$$

$$x = 5$$

$$y = 200000$$

Similar Qns to HW1

An individual has utility over consumption c and leisure l :

$$U(c, l) = c^{1/2}l^{1/2}$$

The individual has total available time T . They can work h hours at wage w , and leisure is given by

$$h + l = T$$

Consumption is financed only by labour income, so

$$c = wh$$

Questions:

- Write the budget constraint in terms of c and l .
 - Formulate the individual's utility maximization problem.
 - Set up the Lagrangian.
 - Find the optimal c^* , l^* , and h^* .
 - Find the Lagrange multiplier λ , and interpret it.
-
-

Solⁿ:

(a) Merge budget constraint

$$h = T - l$$

$$C = \omega(T - l)$$

(b) $\max L(c, l) \text{ s.t. } c = \omega(T - l)$

(c) $c - \omega T + \omega l$

$$L: c^{\frac{1}{2}} l^{\frac{1}{2}} - \lambda(c - \omega(T - l))$$

(d) FOC

$$L'_c = c^{-\frac{1}{2}} l^{\frac{1}{2}} - \lambda = 0$$

$$L'_l = c^{\frac{1}{2}} l^{-\frac{1}{2}} - \lambda(\omega) = 0$$

$$\frac{l}{c} = \frac{1}{\omega}$$

$$l = \frac{c}{\omega}$$

$$c = \omega T - \omega l$$

$$c = \omega T - c$$

$$2c = \omega T$$

$$c^* = \frac{\omega T}{2} \quad l^* = \frac{T}{2}$$

$$(c) \lambda = \left(\frac{\omega T}{2}\right)^{-\frac{1}{2}} \left(\frac{T}{2}\right)^{\frac{1}{2}}$$

λ is the marginal utility that increases if the person gets one more dollar of income.

Marshallian Demand:

A consumer has utility function

$$U(x, y) = x^{1/2}y^{1/2}$$

where $x, y > 0$. Prices are $p_x > 0, p_y > 0$, and income is $m > 0$.

The consumer solves:

$$\max_{x, y} x^{1/2}y^{1/2}$$

subject to

$$p_x x + p_y y = m$$

1. Set up the Lagrangian.
2. Derive the first-order conditions.
3. Solve for Marshallian demand:

$$x^M(p_x, p_y, m), \quad y^M(p_x, p_y, m)$$

$$1. L: x^{\frac{1}{2}} y^{\frac{1}{2}} - \lambda (p_x x + p_y y - m)$$

2. FOC

$$L'_x: x^{-\frac{1}{2}} y^{\frac{1}{2}} - \lambda p_x = 0$$

$$L'_y: x^{\frac{1}{2}} y^{-\frac{1}{2}} - \lambda p_y = 0$$

$$L'_\lambda: p_x x + p_y y = m$$

3. Solve

$$\frac{y}{x} = \frac{p_x}{p_y}$$

$$y = \frac{p_x}{p_y} x$$

$$p_x x + p_y \left(\frac{p_x}{p_y} \right) x = m$$

$$2 p_x x = m$$

$$x^* = \frac{m}{2 p_x}$$

$$y^* = \frac{m}{2 p_y}$$

Hicksian Demand:

Now suppose the consumer wants to achieve utility level $\bar{U} > 0$ at minimum cost.

The consumer solves:

$$\min_{x,y} p_x x + p_y y$$

subject to

$$x^{1/2} y^{1/2} = \bar{U}$$

1. Set up the expenditure minimization Lagrangian.
2. Derive the first-order conditions.
3. Solve for Hicksian demand:

$$x^H(p_x, p_y, \bar{U}), \quad y^H(p_x, p_y, \bar{U})$$

$$(1) p_x x + p_y y - \lambda (x^{\frac{1}{2}} y^{\frac{1}{2}} - \bar{U})$$

(2) FOC:

$$L'_x: p_x - \lambda x^{-\frac{1}{2}} y^{\frac{1}{2}} = 0$$

$$L'_y: p_y - \lambda x^{\frac{1}{2}} y^{-\frac{1}{2}} = 0$$

$$L'_\lambda: x^{\frac{1}{2}} y^{\frac{1}{2}} - \bar{U}$$

(3) Solve

$$\frac{p_x}{p_y} = \frac{y}{x}$$

$$y = \frac{p_x}{p_y} x$$

$$x^{\frac{1}{2}} \left(\frac{p_x}{p_y} x \right)^{\frac{1}{2}} = \bar{U}$$

$$\left(\frac{p_x}{p_y} \right)^{\frac{1}{2}} x = \bar{U}$$

$$x^* = \frac{\bar{y}}{\left(\frac{p_x}{p_y}\right)^{\frac{1}{2}}}$$
$$= \bar{y} \left(\frac{p_y}{p_x}\right)^{\frac{1}{2}}$$

$$y^* = \bar{x} \left(\frac{p_x}{p_y}\right)^{\frac{1}{2}}$$

Compare both

1. Marshallian Demand

vs

2. Hicksian Demand